Exp No : 2

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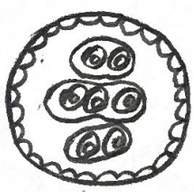
STUDY OF NETWORK CABLES

AIM : study of different types of Network cables.

a) Understanding different types of network cable.

Different types of cables used in networking are :

1. Unshielded Twisted pair (UTP) cable
2. shielded Twisted pair (STP) cable.
3. Coaxial cable
4. Fibre Optic cable.

Cable Type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/ Use	Image
UTP	category 3	10 bps	Advantages: *cheaper in cost *easy to install as they have a small overall diameter Disadvantages: *more prone to (EMI) and noise.	10 Base-T Ethernet Fast ethernet Gigabit ether net Fast ethernet Gigabit ethernet	
	category 5	Up to 100 Mbps			
	category 5e	1 Gbps			
STP	category 6,6a	10 Gbps	Advantages: *shielded *Faster than UTP *less susceptible to noise and interferance. Disadvantages: *expensive *Greater installation effort.	Gigabit ethernet, 10G ethernet (55m) widely used in data centres. Gigabit Ethernet, 10G ethernet (100m)	
SSTP	category 7	10 Gbps			

Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages: *High bandwidth *immune to interference *Low loss bandwidth *Versatile Disadvantages: *Limited distance *cost *size is bulky.	speed of signal is 500m Television network high speed internet connections.
fibre optics cable	Single mode, multi mode	100 Gbps	Advantages: *high speed *high bandwidth *high security *long distance Disadvantages *Expensive *Requires skilled installers.	*maximum distance of fibre optics cable is around 100 metres.



